You mean AI could be wrong?

A Dynamic UI for Appropriate Reliance on AI

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Abstract

People increasingly make decisions with the help of large language models (LLMs), yet they rely on this advice inappropriately—following the AI when it is wrong (over-reliance) and rejecting it when it is right (under-reliance). Most interventions are shown to every user on every trial, regardless of whether that user is currently over- or under-relying on the AI. We argue that the right intervention should depend on the user's current state, which can be read from a single number they can report: their expected AI accuracy. We built DynamicUI, an interface that anchors this number to the AI's true accuracy and, each trial, shows a supporting explanation when the user under-relies, a counter-explanation when they over-rely, and nothing when calibrated. In a small between-subjects pilot ($N = 11$), DynamicUI's total inappropriate reliance is numerically lower than a verdict-only Baseline but not significantly so (Baseline 16.0% vs. DynamicUI 14.2%, n.s.); counter-explanations nonetheless make users hesitate and reconsider, and—importantly for measurement—reported trust and expected accuracy dissociate ($r = 0.45$). We contribute a *capability-anchored trigger* that replaces arbitrary Likert-trust thresholds with a principled, accuracy-relative one; evidence that trust and expected accuracy are *distinct constructs* future studies should measure separately; and a study design that separates an intervention's *attention* effect from its *decision* effect.

Keywords

trust calibration, appropriate reliance, AI-assisted decision-making, adaptive user interfaces, human–AI interaction

1 Introduction

Trust calibration in LLM-assisted decision making is now a central HCI problem: roughly 60 papers document systematic over-reliance on AI output [17]. The effect is large and general—when an AI was only 50% accurate, people working with it were *less* accurate than people working alone (63.9% vs. 74.2%, $p < .01$, $N = 404$) [11]—and people are typically unaware they are being influenced [8].

Table 1: The user viewed as a binary classifier detecting "the AI is wrong." Appropriate reliance means raising Correct Rejection without inducing Under-reliance.

	Human accepts AI	Human rejects AI
AI wrong	Over-reliance (FN)	Correct Rejection (TP)
AI correct	Correct Acceptance (TN)	Under-reliance (FP)

The core difficulty is that users cannot tell *when* the AI is wrong. We treat the user as a binary classifier detecting "the AI is wrong" (Table 1). The goal of appropriate reliance is to raise *Correct Rejection* (catching wrong AI) without paying for it in *Under-reliance* (rejecting correct AI). Existing passive interventions tend to raise Correct Acceptance but leave Correct Rejection roughly unchanged—e.g., explanations raise appropriate AI-reliance yet move Correct Rejection only from 71.9% to 69.5% ($p = .54$, null) [21], and confidence displays calibrate belief but barely shift behavior [20, 25]; transparency can even *reduce* users' ability to catch the model's mistakes [19].

A single static intervention is shown to a user who is over-reliant and to a user who is under-reliant alike—yet these two users need *opposite* nudges [24]. The missing ingredient is a cheap, per-trial read of which state the user is in.

We introduce DynamicUI: an interface that personalizes the intervention to the user's current calibration state, read from their self-reported expected AI accuracy. Anchoring the trigger to the AI's true accuracy (here 70%) defines a *calibrated band* (expected 61–79); at or below it (expected ≤ 60) a supporting explanation restores warranted trust; at or above it (expected ≥ 80) a counter-explanation interrupts over-reliance (Figure 1).

We compare DynamicUI to a verdict-only Baseline in a between-subjects study on two-option ARC science questions [7]. In an underpowered pilot ($N = 11$), the difference in total inappropriate reliance does not reach significance, but two findings do hold up under testing: counter-explanations induce measurable reconsideration, and trust and expected accuracy are not interchangeable measures. Because this pilot is underpowered, we treat behavioral reliance results as preliminary and focus on what the design reveals about calibration measurement and the attention–decision gap.

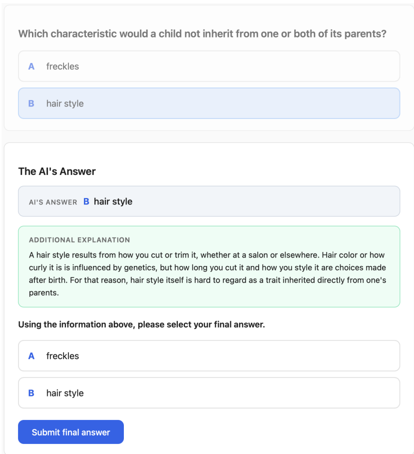
In summary, this paper makes three contributions:

(1) A *capability-anchored trigger*—expected AI accuracy vs. the AI's true accuracy—that operationalizes the textbook definition of trust calibration ("knowing what the AI can actually do") and adapts automatically across AI accuracy levels.
(2) DynamicUI, an interface that maps that trigger to direction-specific interventions (supporting vs. counter-explanation), with preliminary behavioral reliance results.
(3) A study design and evidence that separates attention from decision effects and shows a measured dissociation between trust and expected accuracy.

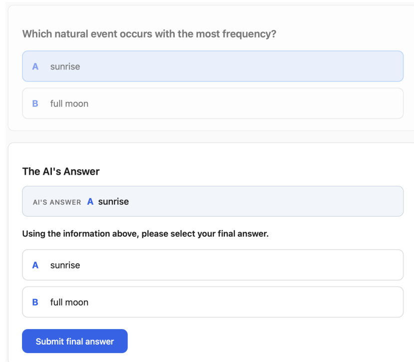
2 Related Work

2.1 Appropriate reliance and its measurement

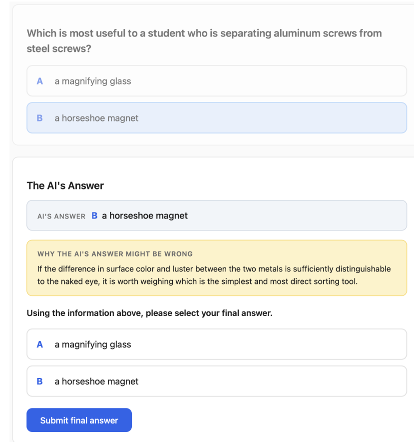
Work on appropriate reliance distinguishes following correct advice from following incorrect advice [2, 21]. We adopt the four-cell confusion-matrix framing (over-reliance, under-reliance, correct



(a) Under-reliance (expected ≤ 60): a supporting explanation ("additional explanation") is added.



(b) Calibrated (expected 61–79): the AI verdict only, with no intervention (= Baseline).



(c) Over-reliance (expected ≥ 80): a counter-explanation ("why the AI's answer might be wrong") is added.

Figure 1: DynamicUI reads the expected AI accuracy the user reported on the previous trial and adapts the current trial's display: (a) a supporting explanation when the user under-relies, (b) the verdict alone when calibrated, or (c) a counter-explanation when the user over-relies.

acceptance, correct rejection) and target *total inappropriate reliance* as the primary outcome.

2.2 Explanations, sources, and confidence—and why they under-deliver

Explanations can reduce over-reliance when they lower the cost of verifying the AI [23], but users often accept the reasoning regardless of whether the AI is right [2, 6], and most never click through to sources (61% never-click, $N = 308$) [12]. Confidence displays move belief but not behavior [20] and fail precisely where confidence contradicts correctness [3, 14]. Unlike these static cues shown to everyone, we show an intervention only when the user's reported calibration warrants it, and we choose its direction from the user's state [24].

2.3 Trust-adaptive interfaces

The closest work, *Adjust for Trust* [22], adapts the UI to the user's Likert trust—supporting explanations when trust is low, counter-explanations when trust is high, and, in a separate experiment, slowing the interaction down (a forced pause) when trust is high—and reports up to 38% lower inappropriate reliance. What motivates our design is its *trigger*: it fires on *Likert trust*, which bundles an affective *trust in the instance* (how the user feels about the AI right now) with *trust in the system* (the user's belief about the AI's overall capability)—two levels that the trust-in-automation literature already distinguishes [10, 13]. We find that these two dissociate (Section 5), so a trust-based trigger may fire on how the user *feels* rather than on a mis-estimate of what the AI can actually do.

We respond on three fronts. (a) We drop AI confidence cues, which an earlier pilot of our own (below) found users largely failed to notice and which prior work finds move belief more than behavior [20]. (b) We replace the arbitrary Likert threshold with an expected-accuracy trigger anchored to the AI's true accuracy, so the calibrated band is principled, portable across domains, and defined on the same scale as the quantity it is meant to track. (c) We deliver the counter-explanation with no enforced pause and measure its effect on *attention* (reconsideration time) separately from its effect on the *decision* (reliance), so a change in one is not mistaken for the other.

This design grew out of an earlier pilot of our own on a FEVEROUS fact-checking task [1]. Against a verdict-plus-source baseline, we tested two ways to lower verification cost [23]: revealing a pre-extracted source snippet in place and replacing numeric AI confidence with a confidence-gated *layout*. We abandoned that pilot because participants treated the untimed task like an exam, checked nearly every claim against its source, and reached near-ceiling accuracy in all arms; most also never noticed the confidence-driven layout. Those failures pushed the present design away from confidence cues and toward an explicit expected-accuracy trigger.

3 Research Questions

- **RQ1.** Without AI confidence cues, does an expected-accuracy–triggered dynamic UI reduce total inappropriate reliance relative to a verdict-only baseline?
- **RQ2 (measurement).** Do self-reported trust (0–10) and expected accuracy (0–100) measure the same construct, or do they dissociate?

Hypotheses.

- **H1.** DynamicUI total inappropriate reliance < Baseline.
- **H2.** In the low subset (expected ≤ 60), DynamicUI under-reliance < Baseline.
- **H3.** In the high subset (expected ≥ 80), DynamicUI over-reliance < Baseline.

H4. If the within-subject correlation between trust and expected accuracy is $r \geq 0.85$, the two are one construct (parsimony); otherwise they dissociate.

Process predictions. DynamicUI trials take longer overall (T1); counter trials take longer than no-intervention trials (T2); counter-explanations lower expected accuracy (M1) and supporting explanations raise it (M2).

4 Method

Table 2: Expected-accuracy trigger zones (true AI accuracy = 70%). $\hat{a}$ is the expected accuracy reported on the previous trial.

Zone	Condition	Intervention
Under-reliance	$\hat{a} \leq 60$	Supporting expl. (3–4 sent.)
Calibrated	$61 \leq \hat{a} \leq 79$	None (= Baseline)
Over-reliance	$\hat{a} \geq 80$	Counter-expl. (1–2 sent.)

Task. Participants answer two-option ARC science multiple-choice questions (Korean) with the help of an AI advisor, in a Sequential Judge–Advisor flow: read the question → give an initial answer and self-confidence → see the AI's verdict (plus any arm-dependent intervention) → give a final answer → see the correct answer → report trust (0–10) and expected AI accuracy (0–100).

Practice trials. The session opens with 5 practice trials—identical in flow and interface to the main block but excluded from all analyses—to absorb *learning effects*. By the time the 20 scored trials begin, participants are already fluent in the task, so the expected-accuracy reading that drives the trigger and the between-arm reliance comparison reflect the manipulation rather than uneven task familiarity.

AI advisor. The AI's accuracy is fixed at 70% by construction—exactly 14 correct + 6 wrong across the 20 main trials (plus 5 practice trials), removing binomial noise and sitting close to the 71% calibrated-AI level used by *Adjust for Trust* [22]. To keep the AI's errors from becoming predictable, the six wrong answers were placed at irregular spacing—the same fixed schedule for every participant—rather than at a single regular interval, so participants were less able to learn a simple pattern (e.g., "the AI is wrong on every fifth question") and anticipate its mistakes in advance.

Conditions (between-subjects). Each participant sees one arm only: (i) *Baseline*—AI verdict only; (ii) *DynamicUI*—verdict + expected-accuracy–triggered intervention.

The trigger. Let $\hat{a}$ be the expected AI accuracy the participant reported on the previous trial. The interface adapts as in Table 2 and Figure 2. The calibrated band is the AI's true accuracy ±10 (70 ± 10), so it moves automatically if the AI's accuracy changes (a 90% AI would make 80–100 the calibrated band). The boundary values themselves ($\hat{a} = 60$ and $\hat{a} = 80$) are assigned to the intervention side rather than the calibrated band, because participants tend to report in round 5–10 increments, leaving the strict calibrated interior at 61–79.

Table 3: Pilot participants ($N = 11$). Anonymized IDs relabeled P1–P11.

Arm	n	Participants
Baseline	5	P1–P5
DynamicUI	6	P6–P11

Why expected accuracy, not Likert trust. Trust calibration is defined as how well a user's trust matches the AI's true capability [24]. Expected accuracy is on the same scale as the AI's true accuracy (both are percentages), so "calibrated" has a principled definition; a Likert threshold (e.g., 5/8) has no such anchor. Trust (0–10) is still collected every trial, but only as a secondary process measure for the dissociation test (H4). The explicit per-trial expected-accuracy question is a study instrument; ordinary deployments would need a less intrusive way to estimate the same quantity.

Why between-subjects. Trust forms and decays asymmetrically (a single salient AI failure drops trust sharply; recovery is gradual) [13, 15], so carryover is not bidirectional and counterbalancing cannot cancel it [9, 18]. A within-subjects participant could also guess the manipulation ("low scores summon cheerleading, high scores summon doubt"), contaminating the trust→reliance signal. Between-subjects blocks both.

Measures. Primary: *Total Inappropriate Reliance* = Over-Reliance (followed a wrong AI) + Under-Reliance (rejected a correct AI); also Switch Rate and Final Decision Accuracy. Process: expected-accuracy trajectory (the trigger), calibration gap $|\hat{a} - 70|$, trust trajectory, and per-stage timing. After the main block, each participant sat for a short semi-structured interview (in Korean) about how they set trust versus expected accuracy and how they read the AI's verdict and any intervention; we use translated excerpts to illustrate—not to test—the quantitative patterns below.

Participants. The pilot reported below collected $N = 11$ (Table 3): Baseline ($n = 5$) and DynamicUI ($n = 6$); each completed 5 practice + 20 main trials.

5 Results

We report descriptive means alongside significance tests: Mann–Whitney U (with Welch's t for reference) for between-arm contrasts, Wilcoxon signed-rank (with paired t) for within-subject contrasts, Brown–Forsythe tests for equality of between-participant variance, and Cohen's d/d_z for effect sizes. With only 5–6 participants per arm these tests are *underpowered*—a null result reflects low power, not the absence of an effect.

5.1 Calibration trajectories

The clearest signal in the pilot is how tightly expected accuracy *clusters across participants*. Figure 3 plots every participant's expected-accuracy trajectory over the 25 trials (5 practice, then 20 main). Both arms open over-confident, with most users near the top of the scale, and then diverge. Under DynamicUI the trajectories *converge*: across the last ten main trials every participant sits in a

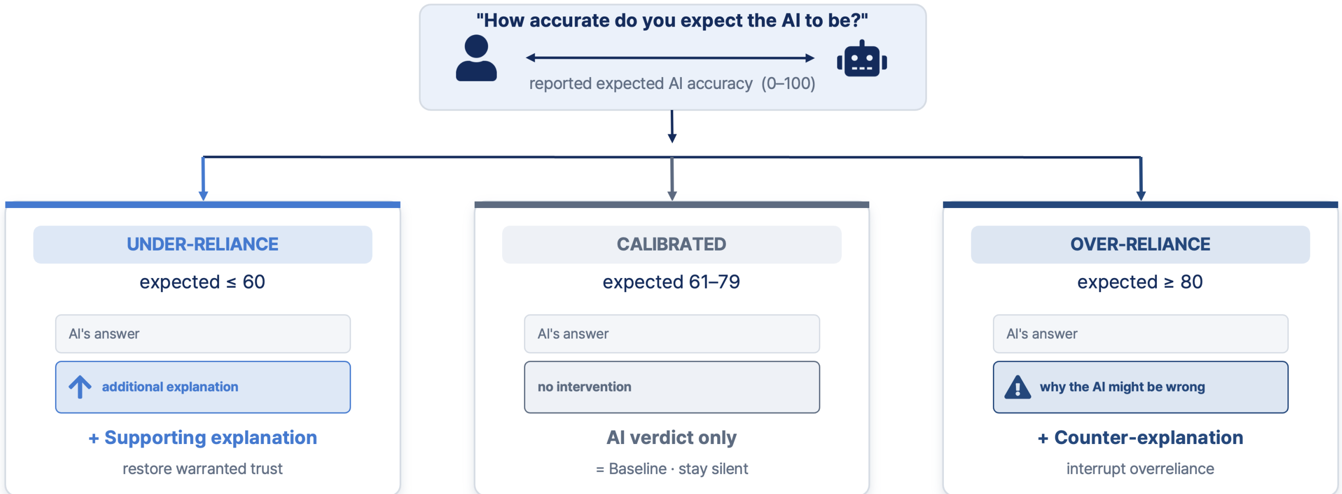


Figure 2: The DynamicUI mechanism: expected accuracy (0–100) routes the next trial's display. *Under-reliance* (expected ≤ 60): the AI's answer is shown with an added supporting explanation, to restore trust. *Calibrated* (expected 61–79): the AI verdict only, with no intervention—identical to Baseline. *Over-reliance* (expected ≥ 80): the AI's answer is shown with an added counter-explanation ("why the AI might be wrong"), to interrupt over-reliance.

Table 4: Hypothesis summary on the pilot ($N = 11$). Between-arm tests are Mann–Whitney U (MWU); within-subject tests are paired t/Wilcoxon. All tests are underpowered, so verdicts are directional: "n.s."=not significant at α=.05. "pp"=percentage points; "pt"= expected-accuracy points (0–100).

#	Prediction	Baseline	DynamicUI	Test (effect, p)	Verdict
H1	Total inappropriate reliance ↓	16.0%	14.2%	MWU p=.40, d=−0.16	**n.s.** (−1.8 pp)
H2	Under-reliance ↓ (low subset)	6.1%	0.0%	denom. too small	untestable
H3	Over-reliance ↓ (high subset)	10.0%	14.0%	denom. too small	n.s. (wrong dir.)
T1	Trial takes longer under DynamicUI	30.4 s	22.1 s	MWU p=.08, d=−0.95	n.s. (opposite)
T2	Counter slower than no-interv. (within)	—	+8.05 s	$t(4)$=2.93, p=.043	**supported** (marginal)
M1	Counter lowers expected (within vs none)	—	+1.7 pt	p=.77	not supported
M2	Supporting raises expected (pooled)	—	+25.8 pt	vs none p=.24 (n=3)	direction only
H4	Trust ≈ expected (r≥0.85)	r=0.74	r=0.25	vs .85: $t(9)$=−3.37, p=.008	**dissociation**

narrow band around the AI's true 70% (per-participant means 69.5–76.0%; between-participant SD 2.8 pt), and 92% of their readings fall inside the 61–79 calibrated band. Under Baseline the trajectories *fan out*: per-participant means span 49.7–77.7% (between-participant SD 13.3 pt; only 60% of readings in band), with some users staying over-confident (near 80%) while others over-correct into under-estimation (≈ 50%).

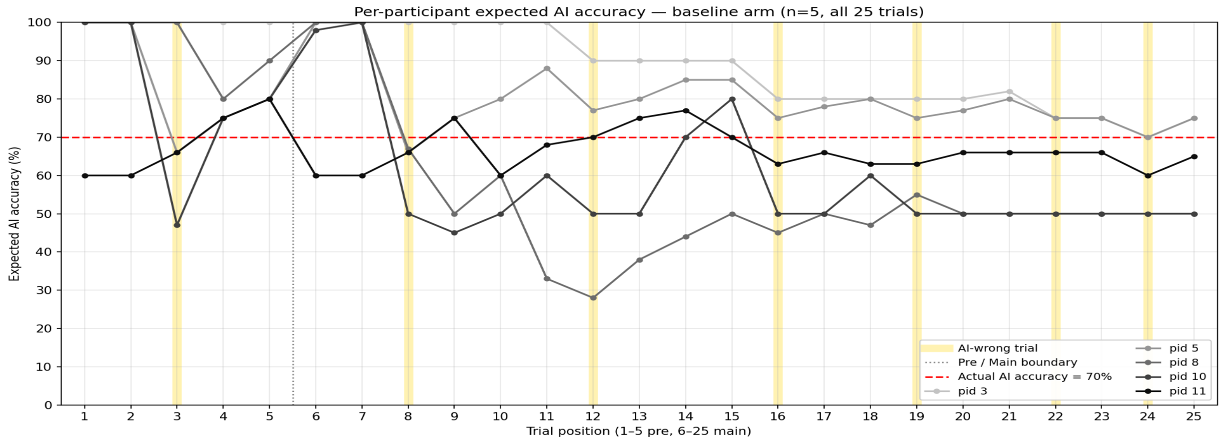
This gap in dispersion is large enough to survive a test despite the sample: between-participant variance is roughly 20× smaller under DynamicUI, and a Brown–Forsythe variance-equality test rejects equal spread (last ten main trials $p = .010$; all twenty $p = .028$). The direction-specific nudges appear to pull each user's belief toward the AI's real capability and hold it there, rather than letting calibration drift idiosyncratically per person—consistent with the trigger steering out-of-band users back toward the band by design. The mean calibration gap $|\hat{a} - 70|$ likewise roughly halves (11.7 → 5.1 pt), though that central-tendency contrast only trends (MWU $p = .067$). We read this as a *belief*-calibration result—DynamicUI converges what users *expect* of the AI—which is distinct from, and (per H1) does not by itself deliver, a change in reliance *behavior*.

A confound is visible directly in Figure 3: in both arms, expected accuracy dips on the shaded AI-wrong trials and rebounds afterward.

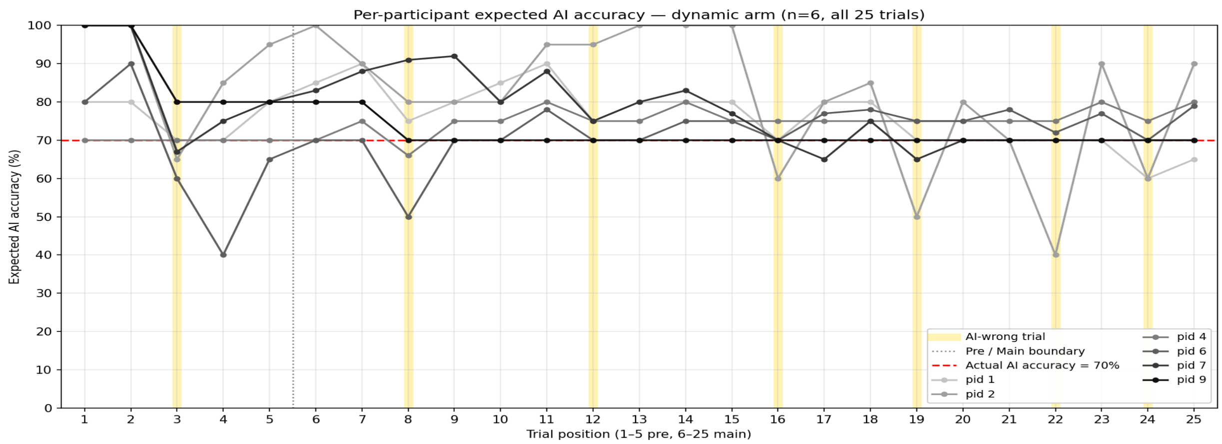
5.2 Hypothesis summary

Table 4 reports each prediction against the pilot data.

What is supported. Two effects survive a significance test despite the sample. (i) *T2*: within DynamicUI, counter trials took +8.05 s longer than no-intervention trials, in the same direction for all five analyzable participants (paired $t(4) = 2.93$, $p = .043$, $d_z = 1.31$; Wilcoxon $p = .063$). Counter-explanations demonstrably capture attention and buy reconsideration time. (ii) *H4*: trust and expected accuracy dissociate (detailed below), the one result that is both significant and not merely a process check.



(a) Baseline (n = 5): trajectories fan out—some users stay near 80%, others over-correct to ≈ 50%.



(b) DynamicUI (n = 6): trajectories converge into a tight band around the true 70%.

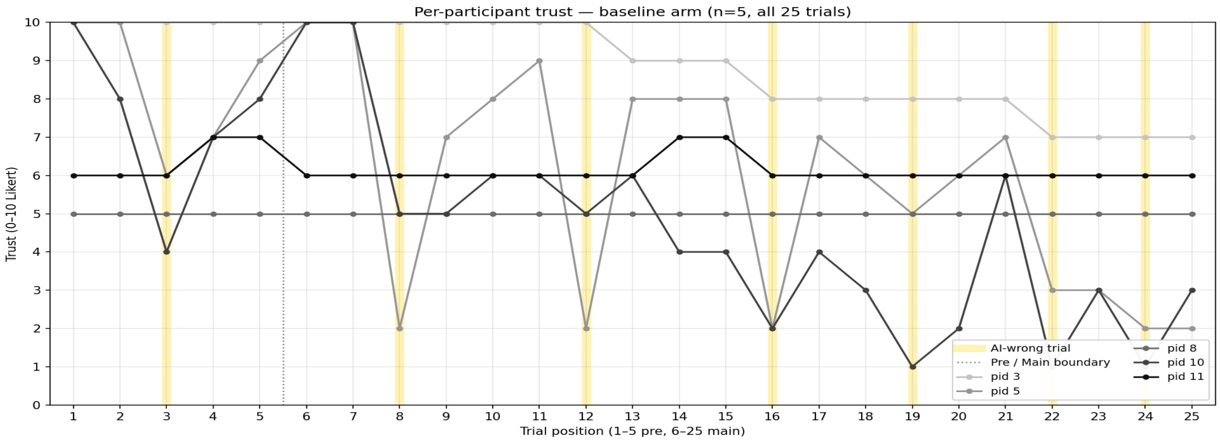
Figure 3: Per-participant expected AI accuracy across all 25 trials (5 practice, then 20 main). Each line is one participant; red dashed = the AI's true 70% accuracy; shaded columns = AI-wrong trials; dotted vertical = practice/main boundary. DynamicUI clusters users near the AI's real capability, whereas Baseline users scatter across a ≈ 50–80% range (between-participant SD 2.8 vs. 13.3 pt over the last ten main trials; Brown–Forsythe p = .010).

What is not supported. The primary outcome *H1* points the right way but is not significant (Baseline 16.0% vs. DynamicUI 14.2%; $p = .40$, $d = -0.16$; 95% CI of the difference [−16.7, +13.1] pp), so we cannot claim DynamicUI reduces inappropriate reliance. *H3* (over-reliance in the high subset) did not drop: counter-explanations slowed people without yet changing the over-reliance *outcome*. *T1* ran opposite to prediction (DynamicUI was faster, ≈ 22 s vs. 30 s; $p = .08$). *M2* moves expected accuracy in the right direction (pooled +25.8 pt), but the stricter within-subject test rests on only three participants and is not significant ($p = .24$).

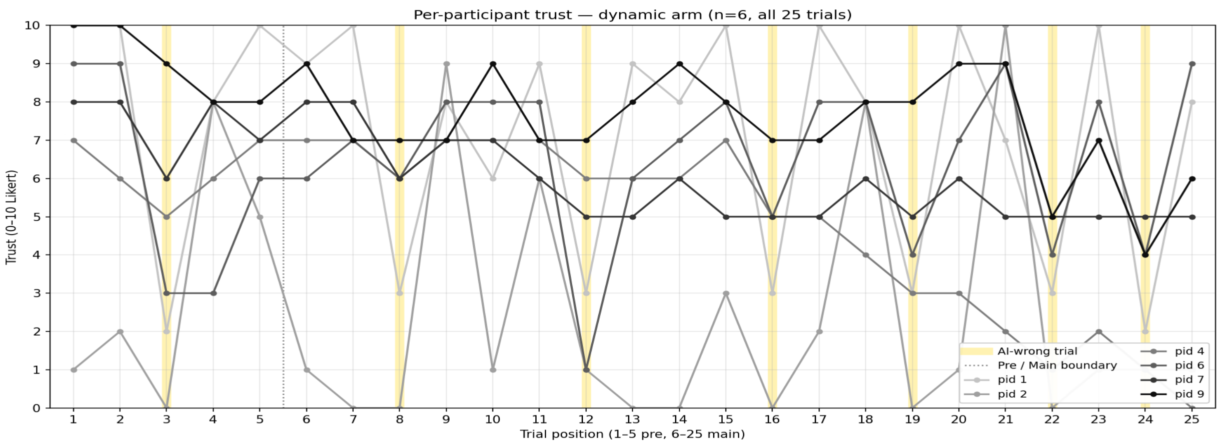
A measurement caveat (M1). Although counter trials are followed by lower expected accuracy on average (−4.10 pt pooled), this does not survive a proper within-subject test (+1.7 pt, $p = .77$). The confound check points to the probe timing: expected accuracy drops mainly when the correct-answer reveal shows the AI was wrong, and same-state comparisons isolate the counter-message effect at only ≈ −1 to +2 pt. We therefore treat M1 as unsupported and flag the probe timing for a design fix.

5.3 Trust trajectory and the dissociation

Figure 4 plots each participant's trust trajectory, and the per-participant view sharpens the contrast with expected accuracy. Baseline trust is comparatively steady—several participants hold a near-constant level across the session—whereas DynamicUI trust is visibly more volatile, swinging down at AI-wrong trials and rebounding (mean trial-to-trial change 2.3 vs. 0.9 Likert points), though with 5–6 participants this trajectory-shape difference is not itself significant (MWU $p = .13$). Crucially, and unlike expected accuracy (Figure 3), trust does *not* converge under DynamicUI: participants stay spread across the scale (per-participant mean trust ≈ 2 to 7) rather than clustering. The two trajectories behave like two different levels of trust [10]: DynamicUI pulls the *system*-level



(a) Baseline (n = 5): trust is comparatively steady—several participants hold a near-constant level.



(b) DynamicUI (n = 6): trust swings sharply at AI-wrong trials and rebounds, and stays spread across the scale.

Figure 4: Per-participant trust (0–10 Likert) across all 25 trials (5 practice, then 20 main). Each line is one participant; shaded columns = AI-wrong trials; dotted vertical = practice/main boundary (no 70% anchor—trust has no validated calibrated value). Unlike expected accuracy (Figure 3), trust does not converge under DynamicUI: it stays spread out and reacts sharply to AI errors.

estimate—expected accuracy, the user's belief about how capable the AI is overall—onto the AI's true capability and holds it there, while the *instance*-level signal—single-trial trust, an affective reaction to the case just seen—stays jumpy and person-specific, tracking each AI error rather than the AI's average reliability.

This dissociation holds up under test. At the within-participant level reported trust and expected accuracy correlate only weakly (mean $r = 0.45$; DynamicUI $r = 0.25$; Baseline $r = 0.74$; only 1 of 10 participants reaches $r \geq 0.85$). The mean correlation is reliably below the $r = 0.85$ single-construct threshold (one-sample $t(9) = -3.37$, $p = .008$) yet reliably above zero ($p = .005$): the two measures are related but not interchangeable—they dissociate [16]. Trust thus carries an affective, instance-level component beyond the system-level capability belief that expected accuracy captures, so the two scales should be measured separately, and a capability-anchored trigger should use expected accuracy, not trust.

6 Discussion

Reframing the contribution. Because the headline reliance difference (H1) is not significant, the contribution is not "a dynamic UI helps a little" but *which part of the loop moves and which does not*, read off the two effects that did survive testing. This reframes an underpowered null into three usable insights.

Insight 1: attention is not decision. Counter-explanations reliably bought reconsideration time (+8.05 s, significant, T2) yet did not, on their own, convert that time into fewer over-reliant choices (H3 null). The *attention* effect of an intervention is separable from its *decision* effect, and winning the first does not guarantee the second. In this respect the counter-explanation behaves much like an AI confidence display, which moves users' belief without moving their behavior [20, 25]: both capture an upstream signal—attention here, stated belief there—without converting it into a different reliance decision. Whether mechanisms that *enforce* deliberation could convert that bought time into a corrected decision remains an open question for future work [5]. The design implication is that capturing attention is necessary but not sufficient—an intervention must also raise the cost of acting without reconsidering. The interviews point to a second reason the bought time seldom converted: the counter-explanation only lands in a narrow band of user knowledge. One DynamicUI participant placed its value in the middle of three tiers—answers one is sure of, answers one has *"heard of but is unsure about"* (the point where *"'reasons it could be wrong' is useful"*), and answers one does not know at all—while another found that for answers already known well it *"wasn't much help."* A Baseline participant, asked what AI help they would want, anticipated the same boundary from the other side: *"if I know nothing about a problem a wrong explanation could pull me along, but if I know even a little I'd catch when the AI explains it wrong."* An intervention fired on the AI's accuracy thus meets a user whose ability to act on it shifts trial to trial with their own knowledge of the item—one more gap between showing a message and changing a decision.

Insight 2: trust and expected accuracy are different instruments. The dissociation (H4) is the pilot's most robust result ($p = .008$). Trust and expected accuracy both react to AI errors, but they track each other only weakly within a person ($r = 0.45$), and they behave differently over the session: under DynamicUI expected accuracy converges on the AI's true capability and holds there, whereas reported trust stays jumpy—swinging down at each AI error and rebounding—and spread across participants (Section 5). This split maps onto the layered view of trust in automation [10, 13], which separates an operator's accumulated, system-level assessment of an automated aid from a dynamic component that fluctuates with the system's moment-to-moment performance. Expected accuracy behaves like *trust in the system*—a comparatively stable estimate of how capable the AI is overall—whereas a single trial's Likert rating behaves like *trust in the instance*—an affective reaction to the case just seen. A Baseline participant drew this line unprompted, calling expected accuracy *"the real metric"* and trust *"whether I'll actually use it or not"*—a capability estimate held apart from a decision to rely. Read this way the dissociation is expected, not anomalous: the two scales sit at different levels of the same construct. The design consequence for a calibration trigger is concrete: it should read the system-level quantity (expected accuracy, on the same scale as the AI's true accuracy), not the instance-level affective signal that Likert trust captures.

Insight 3: outcome feedback moves belief. The M1 confound shows expected accuracy is driven by the ground-truth reveal (≈ -15 pt on AI-wrong trials), not the counter-explanation text (≈ -1 to $+2$ pt). When outcome feedback is present and salient, a short explanation has little additional room to move stated belief. This is both a measurement lesson (the expected-accuracy probe must be decoupled from the correct-answer reveal) and a substantive one (interventions may act on *behavior under uncertainty*, before the answer is known, rather than on post-hoc belief).

Designed for human–AI interaction. DynamicUI is built around a human–AI interaction principle: the same advice needs different scaffolding depending on the user's current trust state. Rather than maximizing information shown, it minimizes it—staying silent in the calibrated band and intervening only at the edges, in the direction the user's own report indicates. The trigger is something the user can actually produce (a single percentage), and it is anchored to the AI's real capability so the system's notion of "calibrated" stays honest as the AI changes.

Implications. The trust–expected-accuracy dissociation is a caution to the broader literature: studies that trigger or measure "calibration" via Likert trust may be capturing instance-level affect as much as system-level capability belief [4, 10]. A same-scale measure (expected accuracy vs. true accuracy) is cleaner both as a trigger and as a dependent measure.

Possible extension beyond the constructed advisor. Because the trigger is anchored to the AI's true accuracy rather than to an arbitrary Likert point, the same idea could be tested wherever a model's capability is estimated. A benchmark accuracy could serve as the anchor a (say 90%), with a calibrated band of half-width δ (say ±5): a supporting explanation would appear when $\hat{a} < a - \delta$ (here < 85), a counter-explanation when $\hat{a} > a + \delta$ (here > 95), and nothing in between. Benchmarks are imperfect proxies for live task accuracy, our advisor was constructed rather than live, and two empirical questions remain open—what band half-width δ counts as "calibrated," and how to infer a user's expected accuracy in ordinary use without prompting for it every turn.

7 Limitations and Future Work

Limitations. (1) *Pilot scale*—$N = 11$, 5–6 per arm; the significance tests we report are severely underpowered, so non-significant results (including the primary H1 contrast) cannot distinguish a small true effect from none, and the supported effects (T2, H4) need replication at the planned sample. (2) *M1 confound*— expected accuracy is reported after the correct answer is revealed, entangling the intervention's effect with the ground-truth reveal. (3) *Explicit elicitation*—asking for expected accuracy on every trial is useful for a study but would be intrusive in ordinary AI use. (4) *Counter-explanation quality*—we do not separate the effect of counter-explanation as a mechanism from the quality, specificity, or persuasiveness of the particular counter-explanations used. (5) *Constructed AI*—a fixed 14/6 schedule controls accuracy but is not a live model. (6) *Narrow task*—two-option Korean ARC science items may not generalize to open-ended or higher-stakes tasks.

Future work. Run the full between-subjects study at the planned sample to power the primary reliance contrast; decouple the expected-accuracy probe from the correct-answer reveal to clean up M1; and broaden stimuli and AI accuracy levels to test whether the auto-adapting calibrated band holds. Future studies should also measure item-level knowledge or familiarity, because the usefulness of a counter-explanation may depend on whether the user knows enough to evaluate it but not enough to decide unaided. Two further directions are prerequisites for the broader extension sketched in Section 6: settling *which band half-width counts as "calibrated"* (we fixed ±10 by fiat, but the proper scale is an empirical question), and *eliciting expected accuracy implicitly*—so the trigger needs neither an explicit per-trial rating nor a study harness and can read a user's expected accuracy during ordinary LLM use.

8 Conclusion

Inappropriate reliance on AI is not one failure but two opposite ones— over-reliance and under-reliance—and they call for opposite nudges. We proposed reading the user's expected AI accuracy, anchoring it to the AI's true accuracy, and letting that single number choose whether and in which direction to intervene. In an underpowered pilot the overall reliance difference was not significant, but two results held up: counter-explanations measurably buy reconsideration time, and trust and expected accuracy are not the same thing—a measurement result worth its own attention. The gap between winning attention and changing the decision is the central question the full study is built to examine.

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This paper used an LLM (Claude Opus 4.7 and Opus 4.8) for literature search, brainstorming the study design, drafting and revising paper, and running the user study analysis scripts. All design decisions, data collection, and claims are the author's.

A Supplementary Materials

All materials below are available in the project repository, https://github.com/Clear-Wisdom/Human-AI-Interaction ; the paths are relative to the repository root.

- **A. Interface & system.** web/ (static SPA; Supabase dynamic_ui_sessions backend; web/config.js thresholds: low = 60, high = 80).
- **B. Stimuli.** arc_curation/ (two-option ARC items, KO/EN) and the 100 pre-generated supporting/counter explanations in web/data/explanations.json.
- **C. Schedule.** Fixed 14-correct/6-wrong main-trial sequence (web/data/schedule.json); the six errors sit at irregularly spaced positions (trials 3, 7, 11, 14, 17, 19), identical across participants.
- **D. Analysis code & data.** Per-participant session JSON (web/backups/: pid_*.json, overall.json) and the analysis and plotting scripts, covering the significance tests (Mann–Whitney/Welch, Wilcoxon/paired t, Cohen's d/d_z, 95% CIs for H1–H4, T1–T2, M1–M2), the M1 confound analysis, and the trajectory plots in web/backups/plots/.
- **E. Full statistics.** Per-participant values and the complete test table are reproduced by the scripts in web/backups/; the planned study will add corrected multiple-comparison reporting at the target sample.

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